MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**Automobile Engineering**	**Course Type**	**PE**
Course Name	**Automotive Communication System**	**Course Code**	**AAE - 4041**
Semester	**VII**	**Max Marks**	**50**
Date	**27.11.2024**	**Duration**	**3 Hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	Explain the importance of Signal-to-Noise Ratio (SNR) in the design of noise-resilient receivers and analyze how understanding thermal and impulse noise contributes to improving their performance.	4	1	1,2,3	1	1,2	L2
1B	Analyze the significance of dBm as a fixed reference level for circuit comparison. Examine the interpretation of negative dBm values, and calculate examples for power levels of 0.01 mW and 0.001 mW.	3	1	1,2,3	1	1,2	L4
1C	Analyze the impact on channel capacity when the bandwidth of a channel increases from 15 kHz to 25 kHz while maintaining a constant SNR of 13 dB.	3	1	1,2,3	1	1,2	L4
2A	Describe the purpose of each part of the Local Interconnect Network (LIN) frame header, including the BREAK field, SYNC pattern field, and identifier field. How do these components contribute to the synchronization and identification processes in LIN communication?	4	2	1,2,3	2	2,3,4,5	L2

2B	Explain the design and purpose of the Controller Area Network (CAN) extended data frame, analyzing its role in node-to-node communication. Support your explanation with a relevant example of its format	**4**	**2**	**1,2,3**	**2**	**2,3,4,5**	**L3**
2C	Describe the mechanism of signal transmission in the Controller Area Network (CAN) two-wire physical layer. Illustrate your explanation with a schematic diagram	**2**	**2**	**1,2,3**	**2**	**2,3,4,5**	**L2**
3A	Describe the role of the Engine Control Unit (ECU) as a vital component in modern vehicles and support your explanation with a relevant diagram.	**4**	**3**	**1,2,3**	**3**	**2,3,4**	**L2**
3B	Discuss the overall flow of the Smart Parking System from vehicle entry to parking space assignment. How do the different technologies interact to automate the process?	**3**	**3**	**1,2,3**	**3**	**2,3,4**	**L3**
3C	Analyze whether a crash sensor with a 10,000 N activation threshold will trigger during a collision. Evaluate the impact force generated when a 1,500 kg car traveling at 40 km/h (11.11 m/s) collides with an immovable barrier and stops completely within 0.08 seconds, and interpret if the force surpasses the sensor's threshold.	**3**	**3**	**1,2,3**	**3**	**2,3,4**	**L4**
4A	Interpret the handoff process in GSM networks, explaining how different network components work together to ensure seamless communication. Illustrate your explanation with a relevant diagram.	**4**	**4**	**1,2,3**	**4**	**1,2**	**L3**
4B	What is the significance of IEEE wireless LAN standards in the automotive sector? Discuss the technical strategies used in IEEE 802.11g to achieve both higher speeds and seamless connectivity with older devices.	**3**	**4**	**1,2,3**	**4**	**1,2**	**L2**

4C	Demonstrate the structure of the Route Reply (RREP) message format in AODV, emphasizing the significance of each field. Explain how these fields facilitate route maintenance and validation in ad hoc networks.	**3**	**4**	**1,2,3**	**4**	**1,2**	**L3**
5A	How do intelligent communication systems in roadways contribute to overall traffic flow efficiency? Interpret how these systems help in reducing congestion and optimizing the movement of vehicles with specific examples.	**4**	**5**	**1,2,3,5,12**	**5**	**1,2,3,18**	**L3**
5B	Interpret how the Topology Broadcast Based on Reverse Path Forwarding (TBRPF) routing protocol enhances communication in vehicular networks. Provide a detailed explanation of its operation and support your interpretation with a relevant diagram.	**4**	**5**	**1,2,3,5,12**	**5**	**1,2,3,18**	**L3**
5C	Bluetooth devices typically have a range of up to 200 meters. The signal strength 'S' of a Bluetooth device decreases with the square of the distance 'd' from the source. Suppose a vehicle's Bluetooth system is designed to function within a range of 100 meters. If the signal strength at a distance of 100 meters is S_0, examine its signal strength at a distance of 200 meters?	**2**	**5**	**1,2,3,5,12**	**5**	**1,2,3,18**	**L4**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	10
2	10
3	10
4	10
5	10

BL	Total Marks
2	17
3	22
4	11
5	